

Prescribed finite-window reaction-time modality by conserved-budget support design

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Can a fixed amount of spatial reactivity be arranged to prescribe the finite-window topology of an encounter-reaction-time density? We answer this existence question for a Doi encounter process. For every fixed finite physical dimension $d \geq 2$, prescribed fixed finite integer m , and compact family of interior catalyst weights, an m -dependent design of ordered narrow catalyst slabs along a monotone exposure trajectory produces exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on a declared compact positive-time window. A global zero bound for the separated Gaussian exposure clocks, a whole-window complement certificate, and compact-positive-time mixed-jet convergence transfer the complete stationary signature to the positive-budget killed process. The construction first fixes a sufficiently small noise-and-width scale and only then takes a sufficiently small installed reactivity budget. It is deliberately nonuniform: the geometry and admissible scales may depend on d and m , and no useful budget threshold, event-mass floor, or stationary-point count outside the declared window is obtained. The theorem therefore establishes prescribed finite modality under a conserved spatial resource without asserting a particular finite-parameter two-dimensional realization.

I. INTRODUCTION

First-passage and reaction-time densities may develop several peaks when distinct trajectory families reach a reactive region on separated time scales [1, 2]. Continuum trap placement can alter capture-time topology [3], and two-particle encounter laws in physical dimensions two and three show related multirate mechanisms [4–6]. Spatially heterogeneous absorption and fixed-total-reactivity optimization are also well established [7–9]. Here we ask a narrower existence question: can the finite-window timing topology of a two-particle reaction be prescribed by support design while the installed amount of static reactivity remains fixed?

The fact that a univariate mixture of m Gaussian components has at most m modes is classical [10, 11]; that upper bound is not our contribution. The issue for an encounter process is to construct every stationary point with a strict margin, exclude additional stationary points throughout a declared time window after the contact factor is included, and carry the complete signature from free exposure to a killed Doi law.

We prove that this can be done for every prescribed fixed finite m in every fixed finite dimension $d \geq 2$. The supports, contact geometry, transport, initial law, and total installed centre-space reactivity are fixed within each constructed family. All interior allocations in that family share the same alternating signature. Thus the result concerns support-geometry design under a conserved budget; it is not a statement that one fixed support family switches among different mode counts when its weights are reallocated.

The Doi killing representation, the associated reactivity-times-occupancy identity, and Gaussian-mixture mode

bounds are prior ingredients [12–15]. The new content is their margin-bearing combination: a separated-clock construction with a global zero count, a uniform whole-window sign certificate, stability under the slowly varying contact probability, and a positive-budget transfer of the full finite stationary signature. The result concerns a transient density under static spatial allocation and is distinct from universal mean-first-passage response to rare activation [16].

II. CONSERVED-REACTIVITY ENCOUNTER MODEL

Let $q(x, t)$ be the unreacted configuration density of two mobile particles, with conservative generator $\mathcal{L}$. Write r for the relative coordinate and c for an encounter-centre coordinate. For a fixed contact profile $\chi_a(r)$ and normalized nonnegative centre-space profiles $\phi_j(c)$, define

$$\begin{aligned} \kappa_{\mathbf{w}}(c) &= B \sum_{j=1}^J w_j \phi_j(c), & w_j &\geq 0, \\ \sum_j w_j &= 1, & \int \phi_j(c) dc &= 1, \\ K_{B,\mathbf{w}}(x) &= \chi_a(r) \kappa_{\mathbf{w}}(c). \end{aligned} \quad (1)$$

Every admissible allocation has the same installed centre-space resource,

$$\int \kappa_{\mathbf{w}}(c) dc = B. \quad (2)$$

This is not an integral over the complete two-particle configuration space. The killed evolution, reaction-time density, and survival probability are

$$\begin{aligned} \partial_t q_{B,\mathbf{w}} &= (\mathcal{L} - K_{B,\mathbf{w}}) q_{B,\mathbf{w}}, & q_{B,\mathbf{w}}(0) &= q_0, \\ f_{B,\mathbf{w}}(t) &= \langle K_{B,\mathbf{w}}, q_{B,\mathbf{w}}(t) \rangle, & -S'_{B,\mathbf{w}}(t) &= f_{B,\mathbf{w}}(t). \end{aligned} \quad (3)$$

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Equation (3) is the usual mass balance for a Doi volume-reaction process, not a novelty claim.

For longitudinal catalyst slabs, translation invariance of the common transverse coordinate gives an exact quotient

$$\Omega_\infty = \mathbb{R}_z \times \mathbb{R}_{r_\parallel} \times \mathbb{T}_W^{d-1}. \quad (4)$$

In the construction below the midpoint Z_t moves monotonically in mean, the relative process remains a positive distance inside the contact region along its deterministic trajectory on the chosen window, and each normalized slab is centred at the midpoint mean at a target time. The $W^{-(d-1)}$ factor in the killing field accounts for the omitted common transverse directions and preserves Eq. (2).

III. POSITIVE-TIME TRANSFER

Write $K_{B,w} = BV_w$ and introduce interior affine control coordinates $\mathbf{w}(\boldsymbol{\theta}) = \bar{\mathbf{w}} + P\boldsymbol{\theta}$. The normalized reaction density and the free-exposure law are

$$F_B(t, \boldsymbol{\theta}) = B^{-1} f_{B,w(\boldsymbol{\theta})}(t), \quad G(t, \boldsymbol{\theta}) = \langle V_{w(\boldsymbol{\theta})}, e^{t\mathcal{L}} q_0 \rangle. \quad (5)$$

On either a bounded reflecting/periodic quotient or the reversible natural-decay realization of Eq. (4), assume q_0 belongs to the corresponding state space and V_w is bounded on a complex control neighbourhood of a compact interior control set. A Dyson expansion followed by time and control Cauchy estimates gives, for every fixed time order r and finite control multi-index α ,

$$\sup_{(t,\boldsymbol{\theta}) \in [\tau, T] \times \Theta} |\partial_t^r \partial_{\boldsymbol{\theta}}^\alpha (F_B - G)| \leq C_{r,\alpha,\tau,T,\Theta} B. \quad (6)$$

The Supplemental Material accompanying this article [17] gives the explicit constant and proof. In particular, C^2 convergence on a compact positive-time window preserves disjoint stationary-point neighbourhoods, curvature signs, endpoint slopes, and a nonzero derivative margin on their complement. The estimate is local in positive time and provides neither a useful numerical value of B nor control on the scale $t = O(B^{-1})$.

A. Prescribed finite modality from separated exposure clocks

The conserved-budget support construction admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this existence result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and mode count and does not assert mode-count switching by weight redistribution on one fixed support.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}. \end{aligned} \quad (7)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), & D_0, \gamma, u_0 &> 0, & z_0 &\neq \bar{z}, \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), & u_0^2 &< 4D_0/\gamma, & \Sigma_{\perp,0} &\succ 0. \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad (8)$$

Here WN is the wrapped Gaussian on $\mathbb{T}_W^{d-1}$, $\Sigma_{\perp,0}$ is a positive-definite $(d-1) \times (d-1)$ covariance matrix, and $|r|_{\text{mi}}$ is the Euclidean norm of the longitudinal relative coordinate together with the shortest transverse representatives modulo W .

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$. Under the minimum-image convention, assume the deterministic relative trajectory $r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$.

Choose target times

$$\begin{aligned} \tau &< t_1 < \dots < t_m < T, & c_j &= x(t_j), \\ x(t) &= \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (9)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{\substack{j=1 \\ w_j \geq w_* > 0}}^m w_j = 1 \right\}, \quad 0 < w_* \leq \frac{1}{m}, \quad (10)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,w,\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (11)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,w,\varepsilon} = BV_{w,\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semigroup and initial law. Then

$$\begin{aligned} G_{\varepsilon,w}(t) &= B^{-1} \mathbb{E}_0[K_{B,w,\varepsilon}(Z_t, R_t)] = \langle V_{w,\varepsilon}, T_0(t) q_0 \rangle, \\ f_{B,\varepsilon,w}(t) &= \mathbb{E}_0 \left[K_{B,w,\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,w,\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (12)$$

The expectation $\mathbb{E}_0$ is with respect to the unkilld process started from Eq. (8). Independence gives the exact factorization

$$G_{\varepsilon, \mathbf{w}}(t) = A_{\varepsilon}(t) H_{\sigma, \mathbf{w}}(x(t)),$$

$$A_{\varepsilon}(t) = \frac{c_{d, \varepsilon}(t)}{W^{d-1} \sqrt{2\pi} \varepsilon \sqrt{D_0/(2\gamma) + \rho^2}}, \quad (13)$$

where $c_{d, \varepsilon}(t) = \mathbb{P}_0(|R_t|_{\text{mi}} < a)$ and $H_{\sigma, \mathbf{w}}$ is defined below.

Theorem 1 (exact prescribed finite modality).— Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock has exactly m nondegenerate maxima and $m-1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint. For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $\mathcal{W}_{w_*}$, such that the budget-rescaled Doi reaction-time density $f_{B, \varepsilon, \mathbf{w}}/B$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B, \varepsilon, \mathbf{w}}$.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$H_{\sigma, \mathbf{w}}(x) = \sum_{j=1}^m w_j \exp \left[-\frac{(x - c_j)^2}{2\sigma^2} \right],$$

$$\sigma = \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}. \quad (14)$$

Writing $q_j(x) = w_j \exp[-(x - c_j)^2/(2\sigma^2)]$, $\pi_j(x) = q_j(x)/H_{\sigma, \mathbf{w}}(x)$, and $L = \partial_x \log H$, let $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the mean and variance of the centres under the finite probability vector $\boldsymbol{\pi}(x)$. Then

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (15)$$

The univariate upper bound of m modes for an m -component Gaussian mixture is classical [10, 11]. Here a direct affine-exponential proof is retained because the Doi transfer needs all critical zeros counted with multiplicity together with uniform root-tube, complement, curvature, and endpoint margins.

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (16)$$

The adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain

uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (15) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (17)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (18)$$

which has at most $2m - 1$ real zeros counted with multiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a posterior-sector sign certificate exclude stationary points on the full box complement. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , compact-positive-time mixed-jet convergence

$$\sup_{\mathbf{w} \in \mathcal{W}_{w_*}} \left\| \frac{f_{B, \varepsilon, \mathbf{w}}}{B} - G_{\varepsilon, \mathbf{w}} \right\|_{C^2(I)} \rightarrow 0 \quad (B \downarrow 0) \quad (19)$$

preserves the disjoint root tubes, signed curvature margins, complement margin, and endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_0(\varepsilon)$. The theorem does not provide a useful uniform lower bound for B_0 , a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical validation gates.

IV. FINITE-PARAMETER SCOPE

The theorem is complete without a discretized example, but its sequential small-parameter construction does not identify a practically resolved physical parameter set. This article includes no numerical evidence for a particular

finite-parameter realization, off-lattice validation, phase diagram, or finite-parameter dimensional comparison. Its scientific conclusions rest only on the exact theorem.

We also make no rigorous numerical claim that a particular finite-volume scheme converges, together with its stationary roots, to the unbounded continuum process. Such a stronger numerical statement would require quantitative consistency, spatial and box-truncation bounds, and margin-preserving transfer of each stationary component. Those additional ingredients are unnecessary for the theorem proved here and do not limit its scope.

V. DISCUSSION

The construction turns catalyst locations into separated exposure clocks. Narrow Gaussian convolution creates one peak near each target time; an affine-exponential zero bound makes the count exhaustive; posterior-sector estimates control the complement after multiplication by the contact probability; and the positive-time estimate

Eq. (6) transfers the strict topology to the Doi law. Together these steps establish arbitrary prescribed fixed finite modality under a conserved centre-space reactivity budget.

The restrictions are substantive. Geometry and admissible scales depend on d and m . The deterministic relative trajectory stays strictly inside the contact set on the design window, and its free contact probability approaches one there as the noise scale vanishes. The limits are sequential, observability at a common useful budget is not guaranteed, and nothing is proved about topology outside the declared window. These limits distinguish the exact existence theorem from a finite-parameter design rule while leaving its finite-window conclusion intact.

DATA AVAILABILITY

This is a purely mathematical work, and no data were created or analyzed.

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